

Course Name: Electrical Circuits (2)
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Date: Jan. 31. 2022
Exam Time: 90 minutes

Prob..	1	2	3	4	5	6	7	8	9	10	11	12	13
Answer													

Prob. 1: Choose the correct answer.

Consider the circuit shown in Figure 1, where the voltage source has $V_s = 5/0^\circ V_{rms}$.

- The open circuit voltage (V_o) between the a-b terminals is
a. $50(1-j1) V$ b. $75(1-j1) V$ c. $25(1-j1) V$ d. None of all
- The short circuit current (I_{sc}) when the a-b terminals are shorted is
a. $-20 A$ b. $-10 A$ c. $-30 A$ d. None of all
- The Thevenin impedance (Z_{th}) between the a-b terminals is
a. $2.5 + j2.5 \Omega$ b. $2.5 - j2.5 \Omega$ c. $-2.5 + j2.5 \Omega$ d. None of all
- The current through a load of $7.5 - j2.5 \Omega$ connected between the a-b terminals is
a. $5(1-j1) A$ b. $10(1-j1) A$ c. $15(1-j1) A$ d. None of all

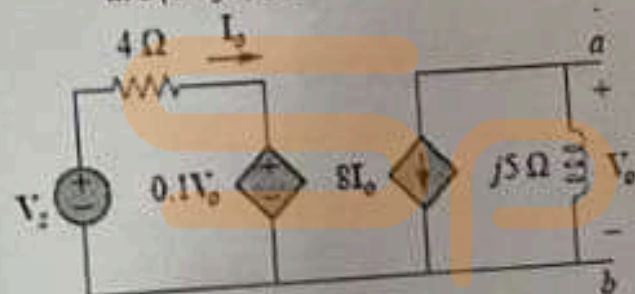


Figure 1

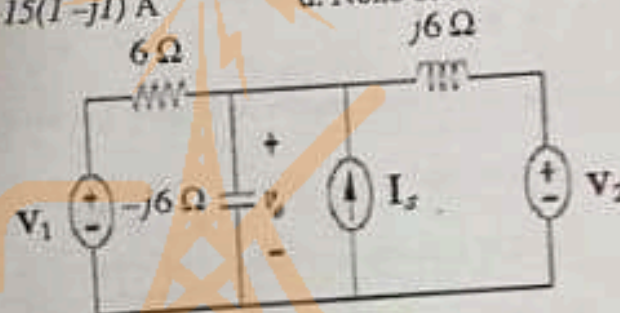


Figure 2

Consider the circuit shown in Figure 2 where $V_1 = 6/0^\circ V_{rms}$, $V_2 = 6/0^\circ V_{rms}$, and $I_s = 1/0^\circ A_{rms}$.

- The voltage across the capacitor due to V_1 source only (in V) is
a. 6 b. 3 c. 12 d. None of all
- The voltage across the capacitor due to V_2 source only (in V) is
a. $-j12$ b. $-j6$ c. $-j3$ d. None of all
- The voltage across the capacitor due to I_s source only (in V) is
a. 6 b. 12 c. 18 d. None of all

Three Loads connected in parallel across a voltage source of $200/0^\circ V_{rms}$, where the loads have:
Load 1: absorbs 250VA at 0.6 lagging p.f., Load 2: absorbs 200W at 0.8 leading p.f., and Load 3: absorbs 300VA and 150VAR.

- The power factor of the load 3 is
a. 0.866 lagging b. 0.707 lagging c. 0.5 lagging d. None of all
- The current of load 1 is
a. $1.3 - j0.75 A$ b. $0.75 - j1.0 A$ c. $1 + j0.75 A$ d. None of all
- The complex power supplied by the source is
a. $564.8 + j501.2 VA$ b. $609.8 + j200 VA$ c. $231.5 + j564.8 VA$ d. None of all
- The power factor of the equivalent load is
a. 0.95 lagging b. 0.748 lagging c. 0.38 lagging d. None of all
- The current supplied by the voltage source in rms is
a. $2.822 - j2.506 A$ b. $2.822 - j1.157 A$ c. $3.049 - j1.0 A$ d. None of all
- The impedance of the equivalent load is
a. $24.87 - j60.67 \Omega$ b. $59.22 + j19.42 \Omega$ c. $60.67 + j24.87 \Omega$ d. None of all

Prob. 1	14	15	16	17	18	19	20	21	22	23	24	25	
Answer													

A balanced Δ -load of $Z_{\Delta} = 8.66 + j5 \Omega$ is supplied by a balanced Y voltages of $V_{as} = 250/\theta^\circ V_{rms}$ in positive sequence.

14. The voltage V_{AB} is
a. $375 + j216.5 V$
b. $216.5 + j375 V$
c. $433 V$
d. None of all
15. The phase current I_{sc} is
a. $37.5 + j21.65 A$
b. $j43.3 A$
c. $-21.65 - j37.5 A$
d. None of all
16. The line current I_L is
a. $37.5 + j65 A$
b. $j75 A$
c. $37.5 - j65 A$
d. None of all
17. The total average power supplied by the source is
a. $48.71 KW$
b. $28.21 KW$
c. $84.37 KW$
d. None of all
18. The total complex power consumed by the load is
a. $56.48 + j50.12 KVA$
b. $56.48 + j23.15 KVA$
c. $48.71 + j28.12 KVA$
d. None of all

Consider a series RLC circuit has $R = 4 \Omega$, $L = 4mH$, $C = 0.4\mu F$, and $V_s = 12/\theta^\circ V_{rms}$

19. The resonant frequency (ω_0) in (rad/sec) is
a. $25 K$
b. $50 K$
c. $20 K$
d. None of all
20. The quality factor (Q) is
a. 50
b. 30
c. 25
d. None of all
21. The cutoff frequencies ω_1 and ω_2 (in rad/sec) are
a. $(19.2, 20.8) K$
b. $(24, 26) K$
c. $(49, 51) K$
d. None of all
22. The voltage across the capacitor (V_C) at resonance (in V) is
a. 400
b. 300
c. 12
d. None of all

Consider the circuit shown in Figure 3, where $V_s = 15/\theta^\circ V_{rms}$, $R_1 = 2 \Omega$, and $R_L = 4 \Omega$.

23. The equivalent input impedance (Z_{in}) seen by the source (in Ω)
a. $4 + j3$
b. $3 + j3$
c. $5 + j3$
d. None of all
24. The current supplied by the source (I_1)
a. $2 - j1.2$
b. $1.6 - j1.2$
c. $2.5 - j2.5$
d. None of all
25. The power absorbed by the R_L resistor is
a. $12.5 W$
b. $10.88 W$
c. $8 W$
d. None of all

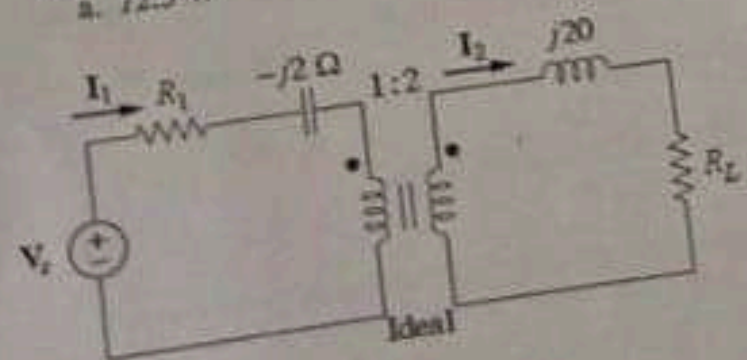


Figure 3

Prob. 1: Choose the correct answer.

1. When the impedance $Z = 8 - j6$ then the voltage across it the current through it by
 a. leads, 36.87° b. lags, 36.87° c. leads, 53.13° d. lags, 53.13°

Consider the circuit shown in Fig. B.1, where $v_s(t) = 150\cos(\omega t + 26.87^\circ)$ and $i(t) = 10\cos(\omega t - 10^\circ)$.

The equivalent impedance at the terminals of the voltage source is
 a. $16 + j12 \Omega$ b. $8 + j6 \Omega$ c. $20 + j15 \Omega$ d. $12 + j9 \Omega$

3. When $\omega = 1000000000000$ rad/s
 a. 15 mH b. 15 μ H c. 9 mH d. 6 mH

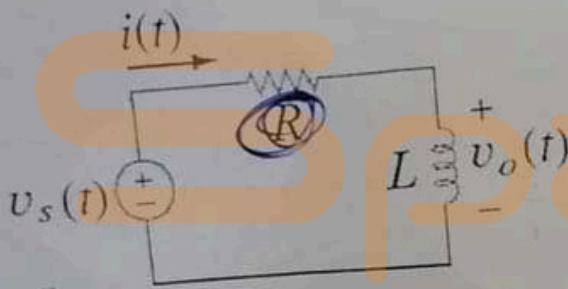


Fig. B.1

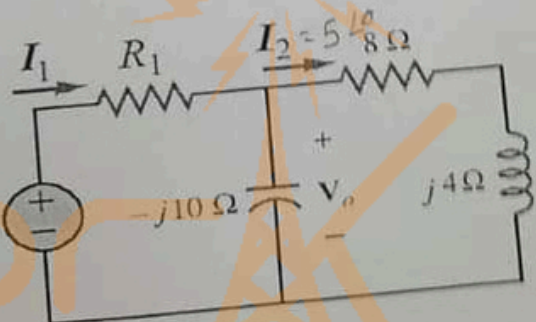


Fig. B.2

Consider the circuit shown in Fig. B.2, where $R_1 = 4 \Omega$.

- The equivalent impedance (Z_{eq}) seen by the source is
 a. $12 - j4 \Omega$ b. $16 - j4 \Omega$ c. $10 - j4 \Omega$ d. $14 - j4 \Omega$
5. When $I_2 = 5/0^\circ$ A then $I_1 =$
 a. $6 + j8$ A b. $3 + j4$ A c. $12 + j16$ A d. $9 + j12$ A
6. When $V_o = 10/0^\circ$ V then $V_s =$
 a. $18 + j4$ V b. $14 + j2$ V c. $16 + j3$ V d. $12 + j1$ V

Prob. 1	1	2	3	4	5	6	7	8	9	10	11	12	13
Answer	a		c				c		c	a	b	c	d

Prob. 1: Choose the correct answer.

1. When the impedance $Z = 8 - j6$ then the voltage across it the current through it by
 a. leads, 36.87° b. lags, 36.87° c. leads, 53.13° d. lags, 53.13°

Consider the circuit shown in Fig. B.1, where $v_s(t) = 150\cos(\omega t + 26.87^\circ)$ and $i(t) = 10\cos(\omega t - 10^\circ)$.

The equivalent impedance at the terminals of the voltage source is
 a. $16 + j12 \Omega$ b. $8 + j6 \Omega$ c. $20 + j15 \Omega$ d. $12 + j9 \Omega$

3. When $\omega = 10000000000 \text{ rad/s}$
 a. 15 mH b. 9 mH c. 9 mH d. 6 mH

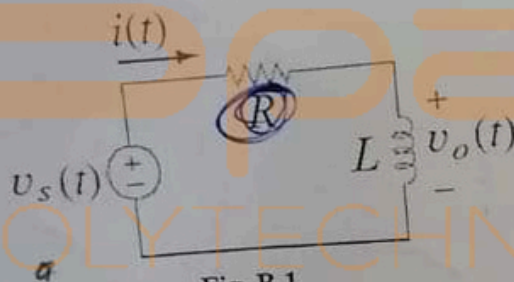


Fig. B.1

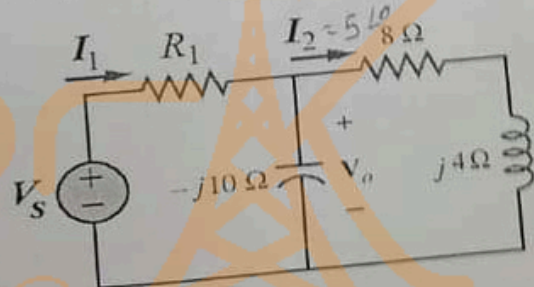


Fig. B.2

Consider the circuit shown in Fig. B.2, where $R_1 = 4 \Omega$.

4. The equivalent impedance (Z_{eq}) seen by the source is
 a. $12 - j4 \Omega$ b. $16 - j4 \Omega$ c. $10 - j4 \Omega$ d. $14 - j4 \Omega$
5. When $I_2 = 5/0^\circ \text{ A}$ then $I_1 =$
 a. $6 + j8 \text{ A}$ b. $3 + j4 \text{ A}$ c. $12 + j16 \text{ A}$ d. $9 + j12 \text{ A}$
6. When $V_o = 10/0^\circ \text{ V}$ then $V_s =$
 a. $18 + j4 \text{ V}$ b. $14 + j2 \text{ V}$ c. $16 + j3 \text{ V}$ d. $12 + j1 \text{ V}$

Consider the circuit shown in Fig. B.3, where $V_1 = 4/0^\circ \text{ V}$ and $V_2 = 3/0^\circ \text{ V}$. (Use nodal method)

7. The voltage across the capacitor (V_x) is
 a. $5 + j1$ V b. $5 + j3$ V c. $5 - j3$ V d. $5 - j1$ V
8. The current (I_1) is
 a. $0.3 + j0.3$ A b. $-0.3 - j0.3$ A c. $0.1 + j0.1$ A d. $-0.1 - j0.1$ A
- The current (I_2) is
 a. $0.05 - j0.15$ A b. $0.1 - j0.05$ A c. $0.15 + j0.05$ A d. $0.2 + j0.15$ A

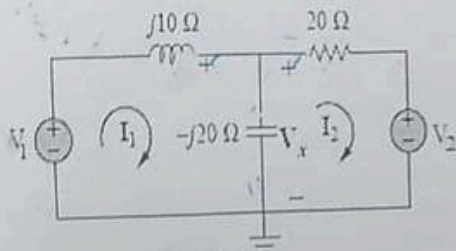


Fig. B.3

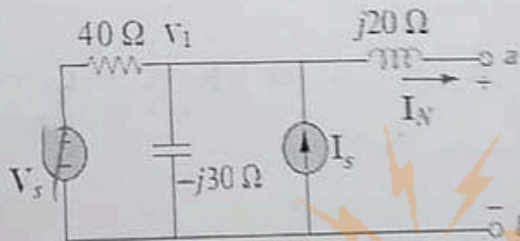
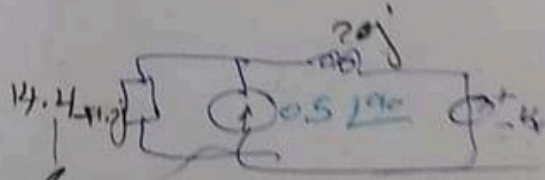
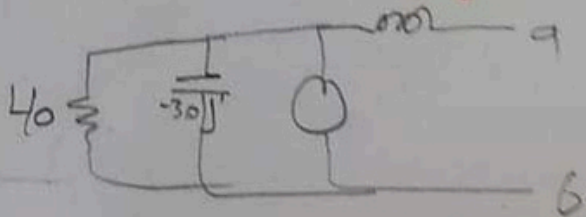


Fig. B.4

Consider the circuit shown in Fig. B.4, where $V_s = 15\angle 0^\circ$ V and $I_s = 0.5\angle 90^\circ$ A.

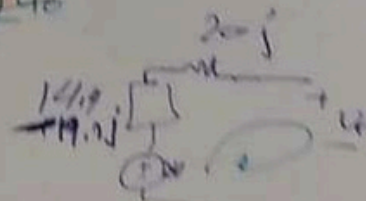
10. The voltage V_x due to the current source only
a. $14.4 + j10.8 \text{ V}$ b. $4.8 + j3.6 \text{ V}$ c. $19.2 + j14.4 \text{ V}$ d. $9.6 + j7.2 \text{ V}$
11. The voltage V_x due to the voltage source only
a. $9 - j12 \text{ V}$ b. $7.2 - j9.6 \text{ V}$ c. $5.4 - j7.2 \text{ V}$ d. $3.6 - j4.8 \text{ V}$
12. The value of V_x due to both sources
a. 15 V b. $42.4 - j12 \text{ V}$ c. $8.4 - j1.2 \text{ V}$ d. $21.6 - j1.2 \text{ V}$
13. The Thevenin equivalent impedance between the a - b terminals is
a. $14.4 + j5.8 \Omega$ b. $40 + j25 \Omega$ c. $j4 \Omega$ d. $14.4 - j19.2 \Omega$



24 $N = I \ell$

~~9.8, 9.2~~

$$-Z_1 + (14.4 + j0.8) I + V_1 = 0$$



Consider the circuit shown in Fig. B.3, where $V_1 = 4/0^\circ$ V and $V_2 = 3/0^\circ$ V. (Use nodal method)

7. The voltage across the capacitor (V_x) is:
 a. $5 + j1$ V b. $5 + j3$ V c. $5 - j3$ V d. $5 - j1$ V
8. The current (I_1) is:
 a. $0.3 + j0.3$ A b. $-0.3 - j0.3$ A c. $0.1 + j0.1$ A d. $-0.1 - j0.1$ A
- The current (I_2) is:
 a. $0.05 - j0.15$ A b. $0.1 - j0.05$ A c. $0.15 + j0.05$ A d. $0.2 + j0.15$ A

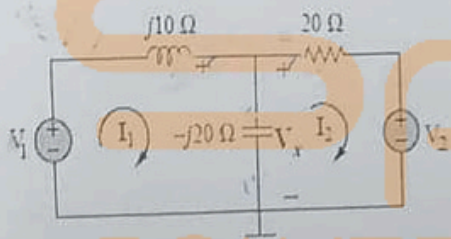


Fig. B.3

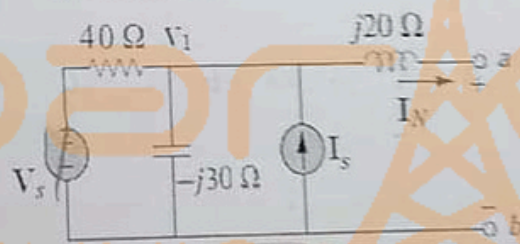


Fig. B.4

Consider the circuit shown in Fig. B.4, where $V_s = 15/0^\circ$ V and $I_s = 0.5/90^\circ$ A.

10. The voltage V_x due to the current source only:
 a. $14.4 + j10.8$ V b. $4.8 + j3.6$ V c. $19.2 + j14.4$ V d. $9.6 + j7.2$ V
11. The voltage V_x due to the voltage source only:
 a. $9 - j12$ V b. $7.2 - j9.6$ V c. $5.4 - j7.2$ V d. $3.6 - j4.8$ V
12. The value of V_x due to both sources:
 a. 15 V b. $42.4 - j12$ V c. $8.4 - j1.2$ V d. $21.6 - j1.2$ V
13. The Thevenin equivalent impedance between the $a-b$ terminals is:
 a. $14.4 + j5.8$ Ω b. $40 + j25$ Ω c. $j4$ Ω d. $14.4 - j19.2$ Ω

- Prob. 1: Consider the circuit shown in Fig. 1.
1. The load impedance (Z_L), in Ω , that absorbs the maximum average power is:
 2. The open circuit voltage between a-b terminals (V_{oc}), when the load impedance is disconnected is:
 3. The short circuit current (I_{sc}), when a-b terminals are shorted is:
 4. The maximum average power transferred to the load impedance (Z_L) is:



Fig.1

(8 marks)

Prob. 2: Consider the circuit shown in Fig. 2 and using mesh analysis method,

1. Write the two mesh equations, do not simplify:

2. The current I_1 is:

3. The current I_2 is:

4. The capacitor current, I_C , is:

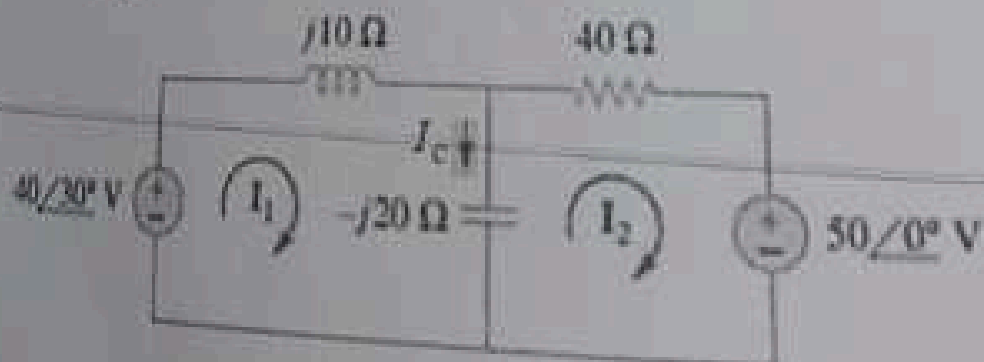


Fig 2

(8 marks)

Prob 3: Two loads are connected in parallel across a voltage source of $100\angle 0^\circ V_{rms}$, where load 1 absorbs $500VAR$ at 0.6 lagging p.f. while load 2 absorbs $400VA$ at 0.75 leading p.f

1. The impedance of load 1 (Z_1), in Ω , is:
2. The current of load 2 (I_2), in Amps, is:
3. The complex power (S) supplied by the source is:
4. The power factor of the equivalent load is:
5. The capacitance needed to raise the pf to unity is:
6. The current supplied by the source, after the power factor is corrected, is:

(12 marks)

Prob. 4: Consider the circuit shown in Fig. 3,

1. The equivalent impedance seen by the source, Z_{eq} , is:
2. The current supplied by the source, I_1 , is:
3. The load current, I_2 , is:
4. The voltage across the primary winding, V_1 , is:
5. The average power absorbed by the 2Ω resistor is:

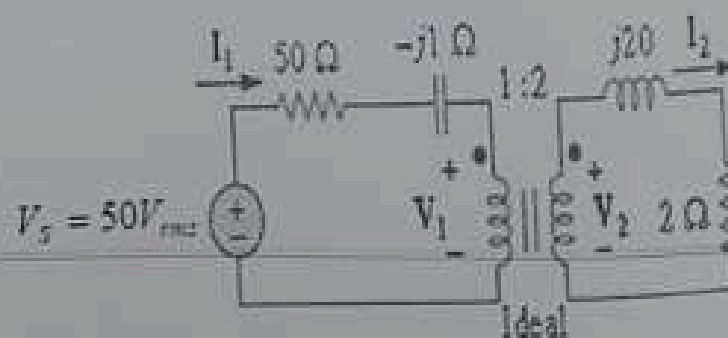


Fig.3

(10 marks)